

SOLUTIONS TO EXAM #2 PRACTICE PROBLEMS

- (1) Prove: If $\lim s_n = 0$ and $s_n > 0$ for all n , then $\lim \frac{1}{s_n} = +\infty$.

Proof. Fix $M > 0$. Since $\lim s_n = 0$, there exists N such that: if $n > N$, then $|s_n - 0| < \frac{1}{M}$.

Assume $n > N$. Then $|s_n - 0| < \frac{1}{M}$. Since $s_n > 0$, then $|s_n - 0| = s_n$. So $s_n < \frac{1}{M}$. So $\frac{1}{s_n} > M$. Since $M > 0$ was arbitrary, $\lim \frac{1}{s_n} = +\infty$. $\square$

- (2) Prove: If $\lim s_n = s$ and (s_{n_k}) is a subsequence, then $\lim s_{n_k} = s$.

Proof. See Theorem 11.3 in Ross. $\square$

- (3) Prove: If (s_n) is a bounded sequence and (s_{n_k}) is a subsequence of (s_n) , then $\liminf s_n \leq \liminf s_{n_k}$.

Solution # 1. Let S be the set of subsequential limits of (s_n) and let S' be the set of subsequential limits of (s_{n_k}) . Since any subsequence of (s_{n_k}) is also a subsequence of (s_n) , we have $S' \subseteq S$. Thus

$$\liminf s_n = \inf S \leq \inf S' = \liminf s_{n_k},$$

where we used the result that the limit inferior of a sequence is equal to the infimum of the set of subsequential limits. $\square$

Solution # 2. Let $A_N = \{s_n : n > N\}$ and $B_N = \{s_{n_k} : k > N\}$. Since (s_n) is bounded, so is (s_{n_k}) . Thus

$$\liminf s_n = \lim_{N \rightarrow \infty} (\inf A_N)$$

and

$$\liminf s_{n_k} = \lim_{N \rightarrow \infty} (\inf B_N).$$

Notice that

$$\begin{aligned} B_N &= \{s_{n_k} : k > N\} = \{s_{n_{N+1}}, s_{n_{N+2}}, \dots\} \\ &\subseteq \{s_n : n > n_{N+1} - 1\} = A_{n_{N+1}-1}. \end{aligned}$$

So $\inf B_N \geq \inf A_{n_{N+1}-1}$. Then

$$\liminf s_{n_k} = \lim_{N \rightarrow \infty} (\inf B_N) \geq \lim_{N \rightarrow \infty} (\inf A_{n_{N+1}-1}) = \lim_{N \rightarrow \infty} (\inf A_N) = \liminf s_n,$$

where in the second to last equality we used the fact that a subsequence of a convergent subsequence converges to the same value. $\square$

- (4) Prove: If (s_n) converges, then (s_n) is Cauchy.

Proof. See Lemma 10.9 in Ross. $\square$

- (5) Prove, directly from the definition of a Cauchy sequence, the following statement: If (s_n) is a Cauchy sequence and there exists a subsequence (s_{n_k}) such that $\lim s_{n_k} = s$, then $\lim s_n = s$.

Proof. Fix $\epsilon > 0$. Since $\lim s_{n_k} = s$, there exists N_1 such that: if $k > N_1$, then $|s_{n_k} - s| < \epsilon/2$. Since (s_n) is Cauchy, there exists N_2 such that: if $n, m > N_2$, then $|s_n - s_m| < \epsilon/2$.

Let $N = N_2$. Assume $n > N$. Fix $k > \max(N_1, N_2)$. Then $n_k \geq k > N_1$ and so $|s_{n_k} - s| < \epsilon/2$. Further, $n_k \geq k > N_2$ and $n > N_2$, so $|s_n - s_{n_k}| < \epsilon/2$. Hence

$$|s_n - s| = |s_n - s_{n_k} + s_{n_k} - s| \leq |s_n - s_{n_k}| + |s_{n_k} - s| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim s_n = s$. □

- (6) Prove using the ϵ/δ definition: $\lim_{x \rightarrow 3} 7x = 21$.

Proof. Fix $\epsilon > 0$. Let $\delta = \epsilon/7$. Assume $0 < |x - 3| < \delta$. Then

$$|7x - 21| = |7(x - 3)| = 7|x - 3| < 7\delta = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \rightarrow 3} 7x = 21$. □